Q1 - 2024 (27 Jan Shift 2)

Considering only the principal values of inverse trigonometric functions, the number of positive real values of x satisfying $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$ is :

(1) More than 2

(2) 1

(3) 2

(4) 0

Q2 - 2024 (29 Jan Shift 2)

Let $\mathrm{x}=\frac{\mathrm{m}}{\mathrm{n}}$ (m, n are co-prime natural numbers) be a solution of the equation $\cos\left(2\sin^{-1}x\right)=\frac{1}{9}$ and let $\alpha, \beta(\alpha>\beta)$ be the roots of the equation $\mathrm{mx}^2-\mathrm{nx}-\mathrm{m}+\mathrm{n}=0$. Then the point (α, β) lies on the line

(1) $3x + 2y = 2$

(2) $5x - 8y = -9$

(3) $3x - 2y = -2$

(4) $5x + 8y = 9$

Q3 - 2024 (31 Jan Shift 1)

For $\alpha, \beta, \gamma \neq 0$. If $\sin^{-1}\alpha+\sin^{-1}\beta+\sin^{-1}\gamma=\pi$ and $(\alpha+\beta+\gamma)(\alpha-\gamma+\beta)=3\alpha\beta$, then γ equal to

(1) $\frac{\sqrt{3}}{2}$

(2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}-1}{2\sqrt{2}}$

(4) $\sqrt{3}$

Q4 - 2024 (31 Jan Shift 2)

If $a=\sin^{-1}(\sin(5))$ and $b=\cos^{-1}(\cos(5))$, then a^2+b^2 is equal to

(1) $4\pi^2+25$

(2) $8\pi^2-40\pi+50$

(3) $4\pi^2 - 20\pi + 50$

(4) 25

Answer Key

Q1 (2) **Q2** (4) **Q3** (1) **Q4** (2)

Q1

$\tan^{-1} x + \tan^{-1} 2x = \frac{\pi}{4}; x > 0$

$\Rightarrow \tan^{-1} 2\mathrm{x} = \frac{\pi}{4} - \tan^{-1} \mathrm{x}$

Taking tan both sides

$$\Rightarrow 2x = \frac{1-x}{1+x}$$

$$\Rightarrow 2x^2 + 3x - 1 = 0$$

$$x = \frac{-3 \pm \sqrt{9+8}}{8} = \frac{-3 \pm \sqrt{17}}{8}$$

Only possible $x = \frac{-3+\sqrt{17}}{8}$

Q2

Assume $\sin^{-1} x = \theta$

$$\cos(2\theta) = \frac{1}{9}$$

$$\sin\theta = \pm\frac{2}{3}$$

as m and n are co-prime natural numbers,

$x = \frac{2}{3}$

i.e. $m = 2, n = 3$

So, the quadratic equation becomes $2x^2 - 3x + 1 = 0$ whose roots are $\alpha = 1, \beta = \frac{1}{2}$ $\left(1, \frac{1}{2}\right)$ lies on

$5x + 8y = 9$

Q3

Let $\sin^{-1}\alpha = \mathrm{A}, \sin^{-1}\beta = \mathrm{B}, \sin^{-1}\gamma = \mathrm{C}$

$\mathrm{A} + \mathrm{B} + \mathrm{C} = \pi$

$(\alpha + \beta)^2 - \gamma^2 = 3\alpha\beta$

$\alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$

$\dfrac{\alpha^2 + \beta^2 - \gamma^2}{2\alpha\beta} = \dfrac{1}{2}$

$\Rightarrow \cos \mathrm{C} = \dfrac{1}{2}$

$\sin \mathrm{C} = \gamma$

$\cos \mathrm{C} = \sqrt{1 - \gamma^2} = \dfrac{1}{2}$

$\gamma = \dfrac{\sqrt{3}}{2}$

Q4

$a = \sin^{-1}(\sin 5) = 5 - 2\pi$

and $b = \cos^{-1}(\cos 5) = 2\pi - 5$

$\therefore a^2 + b^2 = (5 - 2\pi)^2 + (2\pi - 5)^2$

$= 8\pi^2 - 40\pi + 50$